

LightOnOCR vs Baidu Unlimited-OCR

Executive Comparison (Condensed)

Executive Summary

Requirement	Recommendation
Fastest OCR throughput	LightOnOCR-2
Long PDF parsing & table-heavy documents	Unlimited-OCR

Head-to-Head Comparison

Feature	LightOnOCR-2	Unlimited-OCR
Model Size	1B	3B MoE
Long Documents	Good	Best-in-Class
Output	Markdown / JSON	Markdown / JSON / HTML

Performance Overview

LightOnOCR-2

- OlmOCR-Bench: $\sim 83.2 \pm 0.9$
- State-of-the-art among 1B OCR models
- Beats Chandra-9B while being $\sim 9\times$ smaller
- Strong on scientific PDFs, tables, math content, handwriting and multi-column layouts

Unlimited-OCR

- OmniDocBench v1.5: $\sim 93.23\%$
- Formula Accuracy: $\sim 92.6\%$
- Table Accuracy: $\sim 90.9\%$
- +6.2 point improvement over DeepSeek-OCR
- Excels in long reports, manuals, cross-page tables and document-level understanding

Accuracy, Speed & VRAM

Area	LightOnOCR-2	Unlimited-OCR
Accuracy	~ 83.2 (OlmOCR)	~ 93.2 (OmniDocBench)
Speed	3.3x faster than Chandra OCR	$\sim 35\%$ faster than DeepSeek-OCR on long sequences
VRAM	Typically $\sim 8-16$ GB	$\sim 7-8.5$ GB reported
Best For	Enterprise OCR, RAG pipelines	Long PDFs and document intelligence

Hardware Requirements

LightOnOCR-2: Primarily SaaS-based. Estimated deployment profile $\sim 8-16$ GB VRAM on a modern GPU.

Unlimited-OCR: Documented to run on ≥ 8 GB VRAM, with real-world usage reported around 7-8.5 GB. Supports consumer GPUs including RTX 4070/4080/4090.

Other Notable Alternatives

- GLM-OCR (highest reported benchmark accuracy)
- PaddleOCR-VL 1.5 (most mature ecosystem)
- FireRed-OCR-2B

- DeepSeek-OCR-2
- Mistral OCR

Final Recommendation

Choose LightOnOCR-2 for enterprise OCR, RAG ingestion, high throughput and low operational overhead. Choose Unlimited-OCR for self-hosted deployments, long PDF processing, cross-page context preservation and table-heavy document understanding. If benchmark accuracy is the primary criterion, shortlist GLM-OCR and PaddleOCR-VL 1.5 alongside both solutions.